

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	14 July 2026
Team ID	
Project Name	Book A Doctor
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Book A Doctor

DOCTOR APPOINTMENT BOOKING SYSTEM – FLOWCHART

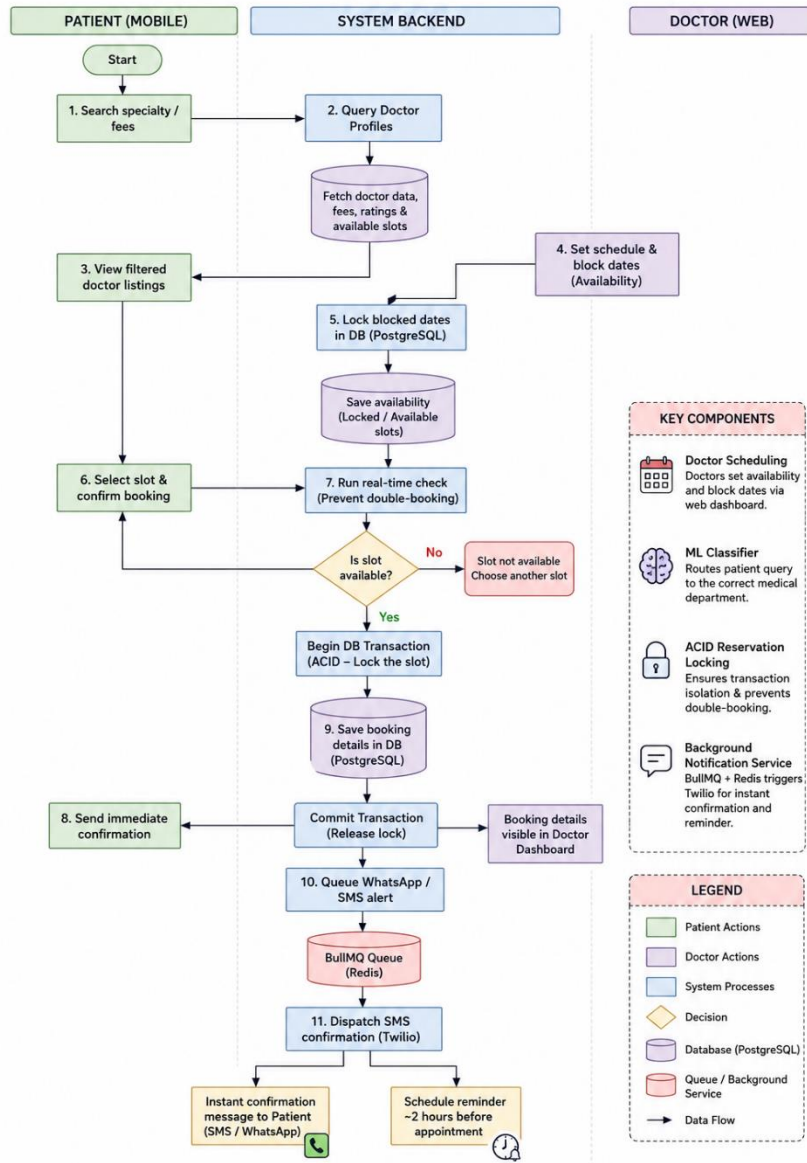


Table-1 : Components & Technologies (Book a Doctor System)

This table outlines the technical specifications and architectural components configured specifically for the Book a Doctor System:

S.No	Component	Description	Technology
1.	User Interface	Responsive Web UI for Doctors and Administrative dashboards; Mobile Native App for Patients to find clinics and book appointments.	React.js, React Native, Tailwind CSS, HTML5, JavaScript
2.	Application Logic-1	Core Backend Engine: Manages CRUD operations for profiles, processes search queries, and runs calendar booking lockouts to prevent double-booking.	Node.js (Express.js) or Python (FastAPI)
3.	Application Logic-2	Telehealth Video consultation engine for secure, virtual real-time doctor-patient checkups.	WebRTC Protocol / Twilio Video SDK
4.	Application Logic-3	Automated Patient Assistant Chatbot: Handles FAQs, schedules basic follow-ups, and guides patients through initial intake forms.	IBM Watson Assistant / Dialogflow

S.No	Component	Description	Technology
5.	Database	Primary Relational Database: Manages strongly consistent ACID-compliant transactional tables (User accounts, doctor schedules, reservations).	PostgreSQL / MySQL
6.	Cloud Database	Fully-managed Cloud Database: Set up with replication across multi-availability zones for failover redundancy and daily snapshots.	Amazon RDS / IBM Db2 on Cloud
7.	File Storage	Cloud-based Object Storage: For secure storing and retrieval of patient medical records, doctor licenses, and generated PDF prescriptions.	Amazon S3 / IBM Cloud Object Storage
8.	External API-1	Communication Gateway: Automatically dispatches SMS, email, and WhatsApp notifications to remind patients of upcoming bookings.	Twilio SMS API / WhatsApp Business API
9.	External API-2	Patient & Doctor Identity Verification Gateway: Integrated with regulatory health registers to authenticate users securely.	ABDM Sandbox API (National Health Identity verification - <i>[Aadhaar Redacted]</i>)
10.	Machine Learning Model	Smart Recommendation Engine: Performs Natural Language Processing (NLP) to recommend a medical specialty based on symptom text entered by patients.	Scikit-learn (Symptom Classifier Model)

S.No	Component	Description	Technology
11.	Infrastructure (Server / Cloud)	Deployment Setup: - Local Server Configuration: Docker containers on developer machines. - Cloud Server Configuration: Container orchestration for scaling.	Kubernetes (EKS), Docker, AWS EC2

Table-2: Application Characteristics:

Table-2: Application Characteristics (Book a Doctor System)

This table outlines the performance, security, and structural qualities configured specifically for the Book a Doctor System:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used to handle cross-platform frontend rendering, server-side APIs, and asynchronous message queues.	Frontend: React.js, React Native (Mobile Framework) Backend: Node.js, Express.js (REST API development)

S.No	Characteristics	Description	Technology
			Background Worker: BullMQ (Queue management)
2.	Security Implementations	List all security controls, data protections, and access points implemented to secure patient medical records and booking histories securely.	Authentication: JWT (JSON Web Tokens) with Auth0 / OAuth 2.0 Data Protection: HTTPS/TLS in-transit, AES-256 encryption at-rest (database & files) OWASP Compliance: Helmet.js (HTTP headers protection), CORS policies, bcrypt (for secure password hashing)
3.	Scalable Architecture	Justify the scalability of architecture (3-tier or microservices deployment).	Microservices Architecture: Decoupled layout allows the Booking Service, Notification Engine, and User Search to scale independently based on demand. Technology: Docker containerization orchestrated via Kubernetes (EKS).

S.No	Characteristics	Description	Technology
4.	Availability	Justify the availability of the application (failover mechanisms, active health checks, and redundant networks).	Redundancy & Load Balancing: Nginx or AWS Application Load Balancer (ALB) dynamically routes traffic across multi-availability zones. Database Failover: Amazon RDS configured with active-standby Multi-AZ automated replication.
5.	Performance	Design considerations optimized for high traffic concurrency, low-latency doctor availability checks, and fast UI page loading.	Caching: Redis in-memory cache to store doctor schedule availability slots and reduce database query overhead. CDN Integration: Cloudflare CDN to cache and distribute static web resources globally.